

Ananya Kulkarni

Engineer | Robotics & Perception Engineer

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Final-year student specializing in multi-modal sensor fusion and SLAM. Thrives in solving low-latency spatial awareness problems for autonomous mobile robots.

Education

B.Tech in Artificial Intelligence, IIT Bombay

CGPA: 8.92 | 2022-2026

Skills

AI & Perception: PointNet++, LiDAR-Visual Odometry, YOLOv10

DevOps: ROS2, Docker, AWS RoboMaker

Work Experience

Autonomous Systems Intern - Skye Air Mobility (2025 - Present)

- Optimized path planning algorithms using A* and DWA, reducing obstacle collision rates by 22%.
- Implemented real-time sensor fusion for IMU and LiDAR data using Extended Kalman Filters.

Projects

Sub-Centimeter Indoor SLAM | C++, ROS2, Factor Graphs

- Achieved 98.5% localization accuracy in GPS-denied environments using GTSAM optimization.
- Deployed on custom-built hardware with NVIDIA Jetson Orin Nano.

Publications

- Fast-SLAM: Real-time Factor Graph Optimization for Edge Devices (IEEE 2025)